

UPI Transaction based Fraud Detection System using Generative AI and Machine Learning Techniques

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Abstract— Growing proliferation of digital payments particularly unsupervised UPI operations in recent years has lead to increased instances of financial frauds worldwide and emphasizes need for a reliable fraud detection framework. Traditional supervised machine learning methods are often limited by inadequacy of fraudulent samples in the dataset and excessive skewness in class distributions. Therefore, this project introduces a fraud detection model with Generative Artificial Intelligence empowered advanced machine learning methods. A Conditional Generative Adversarial Network (CGAN) is used to produce authentic fake fraudulent samples based on transaction behaviours in similar context, facilitates enhancement of training data in a balanced manner, trainest robust classification model using XGBoost to effectively identify fraudulent candidates while NGBoost estimates fraud probability and prediction uncertainties. Also, model transparency is ensured by employing shining-ing (SHAP) views. The system is tested on IEEE-CIS Fraud Detection Dataset of 20000 transactions, with roughly 10% of frauds and results suggest that the model performs markedly better than typical models, which do not seem to incorporate fake generated data information. It attains high fraud detection rate of 84.4% with factually correctness ratio of 90.6% and an F1 measure of 87.4%, a strong indicator of its competence in fraud detection. To summarize, the model boost detection robustness and false negatives, and yields interpretable, principled and cratable predictions. These inform features make it applicable in real-time digital payment scenarios such as UPI and other online financial transactions.

Keywords- Fraud Detection, UPI Transactions, Generative AI, CGAN, XGBoost, Imbalanced Data, Real-Time Monitoring, Anomaly Detection, FinTech Security, Predictive Modeling.

I. INTRODUCTION

In today's world the way people make payments has changed a lot with the use of real-time digital payment systems like Unified Payments Interface. Unified Payments Interface has become one of the popular ways to make payments because it is fast easy to use and accessible to everyone. How ever with more people using payments there has been a big increase in fake transactions, which is a big problem for banks and users. Old systems that detect fraud mostly use rules and basic machine learning. These methods work to some extent. They often miss complex fraud patterns that are always changing. One big problem with detecting fraud is that there are few fake transactions in the data so the machine learning models get better at finding real transactions but not good at finding fake ones. This means they are not very good at detecting fraud especially when it comes to finding all the transactions.

To solve these problems new methods like Generative Adversarial Networks have been developed. Conditional Generative Adversarial Networks are especially useful because they can create examples of fraud based on what we know about transactions. This helps make sure

the system is good at finding transactions. Also new systems need to be transparent and explainable so people can trust them. This project is about making an Unified Payments Interface fraud detection system that uses Generative AI and machine learning. The system uses Conditional Generative Adversarial Networks to balance the data, XGBoost to classify transactions, NGBoost to estimate the risk of fraud and SHAP to explain how the system works. By combining these methods the system is good at finding transactions, accurate and easy to understand. The system provides a solution for detecting fake transactions in digital payment systems, like Unified Payments Interface.

II. LITERATURE SURVEY

With the rapid development of electronic payment and financial technology, fraud detection is always one of the research hotspots in recent years. Imbalanced dataset of real and fraud transactions may affect the precision of a model, which makes others focus on the generation of fraud samples. Some studies are introduced to generate better fraud samples by using generative models. For example, an improved VAE-GAN model [1] addresses the issue of generating more realistic samples by merging Variational Autoencoders with Generative Adversarial Networks, which benefits from the better feature learning for generating synthetic fraud samples. Similar to these mechanisms, GAN-based data augmentation [2] has been proposed and proven to be able to generate needs of fraud samples and accelerate the training of classifiers with improved accuracy.

Conditional GANs (CGAN) take this a step further and condition other relevant information onto the fake samples such as transaction amount, transaction time, merchant category [3]. Conditionally using this information in generation helps the fake samples to be more accurate and specific to the scenario, which enhances its effectiveness during training. Not only that, but CGANs are known to enhance sample quality and perform better than a traditional GAN or a SMOTE method. Semi-supervised GAN [4] not only takes into account the labeled data, but also takes into account the unlabelled data. This method is particularly advantageous when the number of fraud labels is small. Furthermore hybrid approaches such as WDAE-GAN [5] combine Denoising Autoencoders with GANs by deepening the data before generation to remove noise, thereby increasing the quality of samples. Attention-based GAN [6] den knock out irrelevant details during image generation by focusing on the most important features for fraud detection and generation of realistic samples.

However, such methods are computationally intensive and require hyper-parameter fine-tuning. In addition to generation models, classification models are also important components of a fraud detection system. Among these, XGBoost [7] has been perceived as the most effective algorithm for dealing with fraud transaction data due to its speed, accuracy and scalability. By using a Gradient boosting method that offers iterative improvement, this method is well suited for big transaction datasets in the real world. However, XGBoost does not tackle class imbalance issue, making data balancing techniques even more significant to use in conjunction with the model. Systems such as Scarff [8] explores the ability of scalable frameworks to use boosting models to achieve high accuracy at scale in fraud detection. NGBoost [9] introduces probabilistic prediction capabilities into the model so that predictions can also be accompanied with uncertainty measurements. Such capabilities are also supported by research [10], which further shows that NGBoost can be used effectively in fraud detection.

Table 1: Existing Approaches

Ref. No.	Authors	Methodology	Drawbacks	Result
1.	VAE-GAN Model (2023)	Hybrid architecture combining Variational Autoencoder (VAE) for latent feature extraction and GAN for generating realistic synthetic fraud samples to handle class imbalance.	High computational complexity; requires hyperparameter tuning.	Higher Accuracy
2.	GAN-based Fraud Detection (2023)	Trains generator–discriminator network to learn fraud data distribution and augment dataset with synthetic samples for improved classification.	Sensitive to training instability; may produce noisy data.	Data stability
3.	CGAN for Fraud Detection (2023)	Conditional GAN uses transaction attributes (amount, time, location) as input conditions to generate context-aware synthetic fraud samples.	Complex training; risk of mode collapse.	Improved precision
4.	Semi-Supervised GAN (2025)	Combines supervised learning with GAN where discriminator performs classification using both labelled and unlabelled transaction data.	Difficult training balance; higher complexity.	Improved recall
5.	WDAE-GAN (2025)	Integrates denoising autoencoder with GAN to remove noise before generating high-quality fraud samples.	Resource intensive; longer training time.	High performance
6.	Attention-based GAN (2024)	Incorporates attention mechanisms within GAN to focus on important transaction features for better synthetic data generation.	Very high computational cost.	High recall

7.	Chen & Guestrin (2016) – XGBoost	Gradient boosting framework that builds ensemble decision trees sequentially using error minimization and regularization.	Requires external imbalance handling.	High accuracy
8.	Scarff Framework (2023)	Ensemble-based fraud detection framework evaluating multiple boosting models for scalability and robustness.	No data balancing techniques used.	Good performance
9.	NGBoost (2020)	Uses natural gradient boosting to estimate full probability distributions instead of point predictions.	Computationally expensive.	Good calibration
10.	NGBoost Fraud Study (2023)	Applies NGBoost on fraud datasets to generate probabilistic outputs for risk-aware decision-making.	High training cost; tuning required.	Improved probability accuracy

III. PROBLEM STATEMENT AND OBJECTIVES OF THE PROJECT

The way people make payments has changed rapidly in recent years, especially with the growth of Unified Payments Interface (UPI). While this has made transactions faster and more convenient, it has also led to a noticeable rise in fraudulent activities that are becoming harder to detect. Most existing fraud detection systems rely on traditional machine learning models or rule-based methods, and these approaches face several challenges. One of the biggest issues is that fraudulent transactions are very few compared to genuine ones. Because of this imbalance, models tend to perform well on normal transactions but struggle to correctly identify fraud. Another concern is the heavy focus on accuracy as a performance measure. A model can appear highly accurate overall, yet still miss many fraudulent cases, which can result in serious financial losses.

There are also limitations related to transparency and decision-making. Many current systems behave like black boxes, making it difficult to understand how a prediction is made, which reduces trust in their outputs. In addition, these systems often do not provide information about how likely a transaction is to be fraudulent, even though such insights are important for taking action. Fraudsters are constantly changing their strategies, which means detection systems must also be flexible and able to adapt over time. For these reasons, there is a need for a more balanced and reliable approach—one that can effectively detect fraud despite limited examples, provide clear explanations for its predictions, and offer a measure of confidence in its results. Such capabilities are essential for modern digital payment environments like UPI.

- To develop a UPI transaction fraud detection system capable of identifying fraudulent activities.
- To address class imbalance in fraud data for synthetic fraud sample generation.
- To implement Classification algorithm that reduces false negatives and improves fraud detection accuracy.
- To incorporate an algorithm for probabilistic fraud risk estimation, enabling risk-aware and confidence-driven transaction decision-making.
- To provide explainable fraud predictions for transparent and trustworthy decision-making.

IV. PROPOSED METHOD

The pipeline for fraud detection is performed by collecting the transaction data, data pre-processing and use of CGAN module to balance the imbalanced data by generating synthetic class minority data and using XGBoost to do classification, using NGBoost for risk prediction and finally using SHAP to explain the results and output the results with fraud statues and risk score. The architecture of the fraud detection system shown in figure 1 is based on transaction dataset architecture proposed in this paper consists of a well-defined and sophisticated pipeline that brings together multiple novel and latest technology APIs into a unified and integrated flow. The overarching concept of this UPI transaction fraud detection system architecture is that it aims to be substantially accurate, trustworthy, open, forward looking, and scalable by efficiently combining the authoritatively dynamic UPI transaction data stream analysis, generative AI, specialized ML algorithms, and the transparency enableible explainable AI in a pipelined workflow.

The first step is the collection of data related to transactions, which are the key of the entire system. It takes data from two sources: from the set of stored data, which in the framework of this project was the IEEE-CIS dataset and from a user, who can provide CSV files and how to simulate transactions. Data are constituted from some parameters such as transaction amount and time, transaction merchant and its place of origin. These parameters are important since it is not usual to make a fraud decision based on one parameter, but in the decision are involved relationships between one or more parameters. For example, a transaction is not necessarily suspicious if alone seem regular, but is not when it is considered together with the transaction time and the origin place of it. This is the first point in order to the system get rich and accurate data. After the data is collected, it is passed to the preprocessing step, which is an essential point of data analysis. Transaction data may contain anomalies, null values, and noise. These factors may have an adverse effect on the performance of the machine learning models. For this purpose, some preprocessings are performed. Null values are estimated using some appropriate techniques (imputation). Some unimportant or redundant features are excluded. Feature extraction is performed, i.e., some new features are estimated based on the existing features.

For example, the system might be able to calculate how often certain features are experienced during that period, e.g., the number of transactions on one day, or the way users behave. The system is also able to perform the conversion of numerical data to categorical data and vice versa, as well as standardization or normalization of numerical data, which are needed because of the requirement of the machine learning algorithms. The empty data structure after preprocessing is well-formed and ensures a high level of high quality and it is understandable for the computer. This step is necessary for all the subsequent processes. The most important factor that makes fraud detection challenging is the already mentioned unbalanced data.

Normally the number of fraud cases in a real dataset is much lower than those of normal transactions.

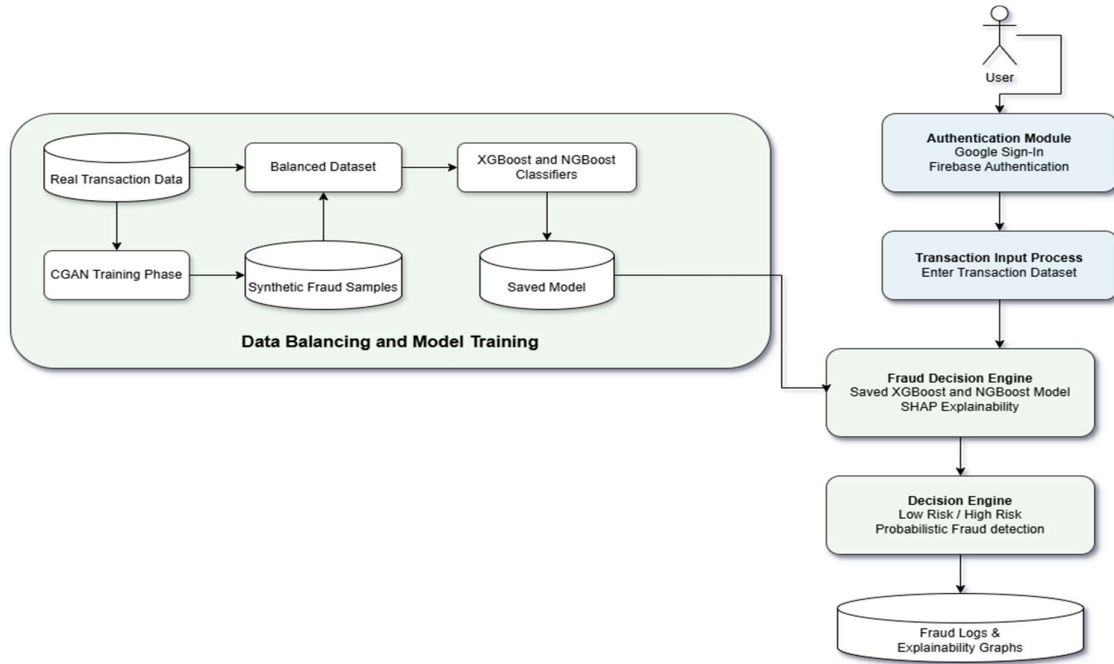


Figure 1: Architecture Diagram

This causes a bias of machine learning algorithms to make a decision that all cases are normal. As a result, the ability to identify fraud decreases. That is why the architecture contains a specific block called Conditional GAN which is able to generate fraud samples. It is a neural network structure, which is formed by two neural networks discriminator and generator working in a competition. The generator uses random numbers and some conditional attributes (e.g., time or amount of transaction) to create artificial data, which is then tried to be classified by the discriminator as real or fabricated. Both networks learn in a competition, i.e., the generator learns how to generate more realistic data, and the discriminator enhances at classifying real or fake data. After some iteration this kind of network yields very good quality synthesized data, which can be used together with the initial data as a clear and well-structured dataset, which is a great input for machine learning algorithms.

Once the data has been balanced, the cleaned data flow into the machine learning layer, which is the core decision maker of this system. In this layer, two very powerful algorithms will be used; XGBoost and NGBoost. The XGBoost engine will determine if a transaction is fraud or not; it is a Gradient boosting algorithm which produce through multiple decision trees, sequentially, a final classification/regression decision, learning from the errors of the previous trees. As a result, it allows learning complex patterns in structured data. While it produces a classification, XGBoost does not give an indication of its certainty level regarding the result. Which is why it will be complemented by the results derived from the NGBoost engine; which extends the capabilities of the XGBoost by adding probabilistic predictions for the outcome, which means producing an estimated likelihood of the transaction being fraud. Instead of

simply outputting the classification for the transaction, it gives a score of how big is the potential risk. Those scores will allow further flexible decisions compared to classical binary rules. Like a transaction with a high level of misclassification can be automatically blocked; while transactions with a moderate likelihood of being fake (based on the scores of the model) can be just flagged for manually checking.

Algorithm of Proposed System:

Input: D: Historical dataset, R: Request

Output: Fraud prediction results / Batch analysis

```
1  Begin
2  Read D
3  if isImbalanced(D) then
4    D ← generateSyntheticData_CGAN(D)
5  end if
6  M1 ← train_XGBoost(D)
7  M2 ← train_NGBoost(D)
8  saveModel(M1), saveModel(M2)
9  initializeServer()
10 loadModel(M1), loadModel(M2)
11 E ← initializeSHAP(M1)
12 while true do
13   R ← receiveRequest()
14   if R.type = "single" then
15     sender ← R.sender_upi
16     receiver ← R.recipient_upi
17     amount ← R.amount
18     category ← lookupRecipient(receiver)
19     F ← generateFeatures(category, amount)
20     F ← preprocess(F)
21     P ← predict(M1, F)
22     if category = "Blacklisted" then
23       P ← max(P, 0.95)
24     end if
25     if P ≥ 0.75 then
```

```

26     risk ← "HIGH"; action ← "BLOCK"
27   else if  $P \geq 0.45$  then
28     risk ← "MEDIUM"; action ← "VERIFY"
29   else
30     risk ← "LOW"; action ← "ALLOW"
31   end if
32   S ← computeSHAP(E, F)
33   explanation ← getTopFeatures(S, 5)
34   sendResponse(P, risk, action, explanation)
35 else if R.type = "batch" then
36   T ← loadCSV(R.file)
37   T ← preprocess(T)
38   P_list ← predictBatch(M1, T)
39   S_list ← computeSHAPBatch(E, T)
40   for i ← 1 to length(T) do
41     if  $P\_list[i] \geq 0.85$  then
42       risk[i] ← "HIGH"
43     else if  $P\_list[i] \geq 0.60$  then
44       risk[i] ← "MEDIUM"
45     else
46       risk[i] ← "LOW"
47     end if
48     explanation[i] ← getTopFeature(S_list[i])
49   end for
50   stats ← computeStatistics(risk)
51   sendBatchResponse(P_list, risk, explanation, stats)
52 end if
53 end while
54 End

```


Module Connectivity:

The module connectivity diagram (Figure 2) displays the interrelation of all the components of the system. It represents flows from the frontend to the backend, then preprocessing, machine learning and generation of results.

The Proposed AI system in Figure 2 is to detect the fake UPI transactions in real-time. It is a collaborative system that working with each component to perform different functions. This system has a user interface that provides users the ability to enter data (or) upload a file that contains a whole bunch of transactions data. It then analyzes the data and predicts whether it is fake (or) real based on its prediction and interprets the reason for such prediction. Detected Frontend component firstly which is user interactive component of this system, built on the React platform. This component has two different functionalities provided to the user: one for simulating fake transactions for testing purpose, and the other to upload a file with transactions data, which is then sent to backend component. Backend component works as brain of this system, which receives data and analyzes it to predict whether the data is fake (or) real, and interprets the reason prediction made.

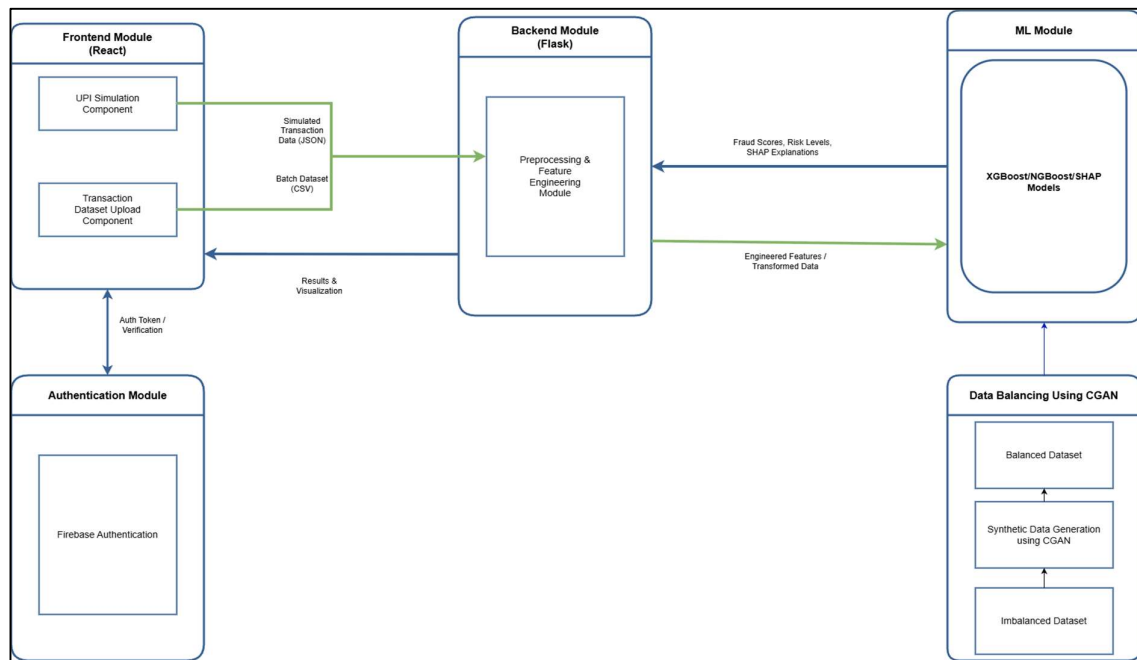


Figure 2: Module connectivity Diagram

The communication occurs between these two components using JSON, CSV, etc. Formats. Have a special authentication module that ensures only authorized users can use the system. This module leverages Firebase Authentication to create users, allow them to log in, and handle their sessions. When a user logs-in they receive a token which must be included in every request they send to the system. This way the system is protected and only authorized users can gain access. The backend is the central part of the system. It is built using Flask. It receives requests from the frontend, validates that they are authentic and passes them to the relevant part of the system. An important part of the backend is the preprocessing module, which receives the raw data from the transactions, cleans it, replaces missing values, and standardizes it before passing it to the machine learning module. Once the data has been pre-processed it is sent to the machine learning module, which is the 'life' of the system. It employs 2 models (XGBoost and NGBoost) for classification of whether a transaction is real or fake. XGBoost is a high-

performance implementation of gradient boosting and it is especially good at detecting regularities within data, while NGBoost is capable of providing not only a classification but also an actual probability to the prediction. Combining the power of both models the system is able to accurately predict the class of a transaction. The system goes beyond the simple binary decision of whether a transaction is fake or not.

V. RESULTS AND DISCUSSION

Description about Dataset

The data set used here is the UK Handing Fraud data set from the IEEE-CIS Fraud Detection Data Set. This data is the standard database that is utilized in the construction and assessment of systems regarding financial fraud detection. It consists of nearly 20000 real-world transaction records, of which some of the activities are legitimate while others are counterfeit. This data set is distinct since the imbalance present means that around 90% of the data is valid, leaving only very little of the data to be forgeries. As this is the case, the machine learning algorithms that are used to train the models have to then learn to have predictions that lean toward the transactions being legitimate, but this leaves untrained the ability to detect fraud. The problem is that there needs to be a specific way in which the data is prepared, such that the model can distinguish between the transactions and decipher whether or not said activity is suspicious. One of the features of each transaction is that it will include additional metadata that can help in contextualizing the commerce. Many factors such as transaction volume, when it was performed, what store it was conducted at, and what location it was carried out at, as well as the behaviour history of the consumer, are crucial to acquiring the inconspicuous differences between what is normal and what is suspicious.

Table 2: Dataset Details

S.NO	Attribute	Description
1	Transaction ID	Unique transaction identifier
2	Transaction Amount	Monetary value
3	Transaction Time	Time of transaction
4	Merchant ID	Merchant identifier
5	Merchant Category	Type of business
6	Location	Transaction location
7	Previous Transactions	Number of past transactions
8	Average Amount	User's average spending
9	International Flag	Domestic / International
10	Fraud Label	Target variable

Detailed explanation about the Experimental Results

Justification of parameters:

In the Proposed System, how to choose parameters of performance is a very important issue. Choosing suitable parameters based on the classification performance would be certainly very crucial since the class imbalance and combined complex detection models for fraud detection problem.

Learning Rate:

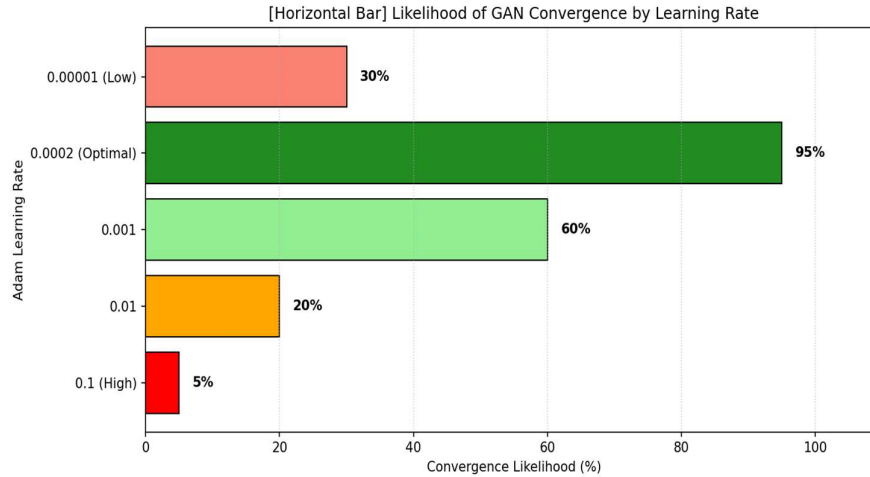


Figure 3: Learning Rate

In the Proposed System the learning rate as in the Figure 3 for the GAN (it is in the particular case the CGAN) is a very constraining parameter for the learning, which influences dramatically the stability of the training. Indeed when using a GAN we are working with two competing networks the Generator and the Discriminator and these two have to be learned at the same rate. If one is learning more quickly than the other then it will be too dominant and the training will not converge.

A very small learning rate (e.g. < 0.0001) makes the training process very slow. Sometimes the Discriminator may not be able to recognize any trend in the data and end up analyzing the noise itself. Therefore, the networks stop learning and the training gets stuck. When the learning rate is very high (e.g. 0.001-0.01) the training becomes unstable. The Generator keeps trying to learn faster so that it can outsmart the Discriminator and make it believe that the false data is actually real. On the other hand, the Discriminator shifts the decision boundary position very fast resulting in the oscillation of the loss value to a very high (overflows to infinity). Hence a middle way (around 0.0002) is chosen, which is the learning rate recommended by many researchers for GAN based models especially CGANs, so that both the networks learn at a reasonable pace without dominating each other. This ensures stable convergence of both networks and hence a good quality of synthetic data

Epochs:

In the Proposed System, the number of epochs determines how many times the model is exposed to the entire dataset during training. For Generative Adversarial Networks, this parameter is especially important because the goal is not simply to minimize loss, but to maintain a balance between the Generator and the Discriminator. As shown in Figure 4 it clearly shows that selection of epochs is indeed very critical for generation of synthetic data that is fairly realistic as well as diversified. When the number of epochs is too low (say below 800), it might not have trained the network fairly enough (the network has not learned the distribution of the dataset). The Generator is not able to understand the complex relationships between different features in the dataset and generates fake data which can easily be identified. When the number of epochs is too high (say above 2500), the network might suffer from “mode collapse” where the Generator has identified a number of ways to fool the Discriminator, and the Generator keeps generating the same data repeatedly.

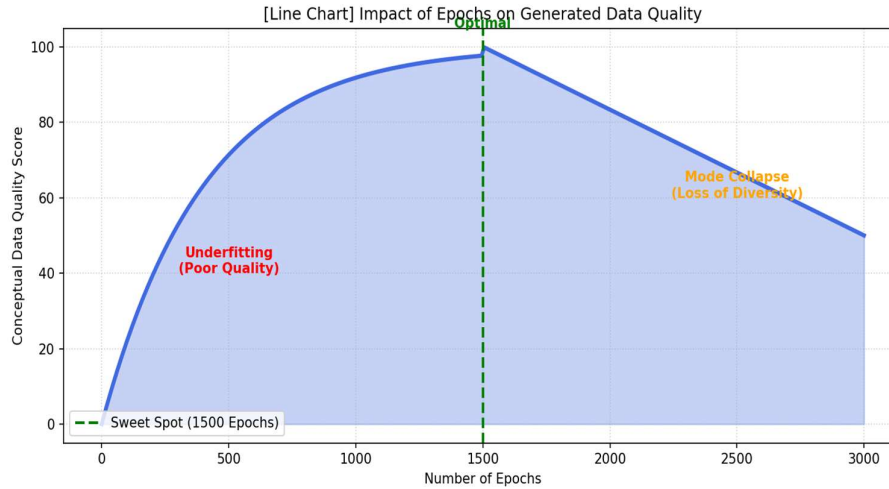


Figure 4: Epochs

In order to attain the best trade-off, approximately 1500 epochs are used in the Proposed System. This is sufficiently long for the GAN to learn the underlying data distribution, but not so long as to lead to mode collapsing. As indicated in Figure 4, the amount of training the generator receives is well balanced, resulting in data that is realistic and is reasonably dispersed. This ultimately increases the performance of the fraud detection model.

Experimental Results:

The comparison of the capability of detecting fraud transaction vs. performance of fraud detection system in our Proposed System are tested in several experiments. The experiments are performed in order to compare various techniques: train the classifier with an imbalanced original dataset, apply GAN to amplify the dataset, and utilize the CGAN method which we propose in this paper.

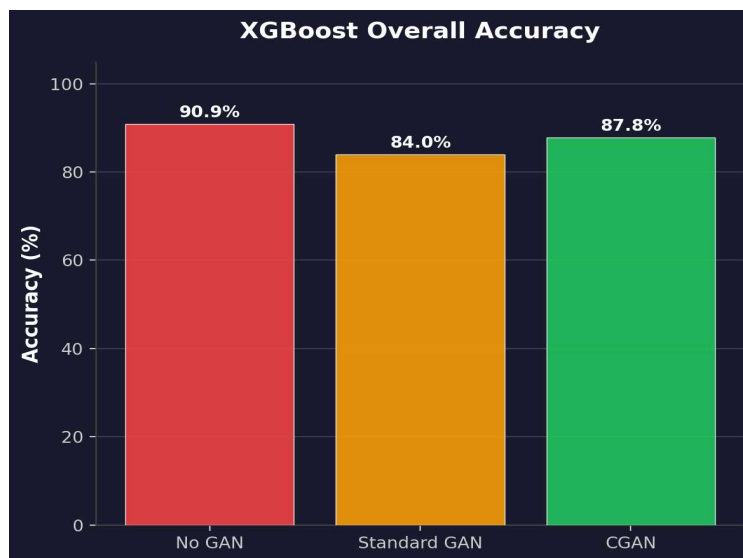


Figure 5: Overall Accuracy

As shown in Figure 5, these comparisons assist in determining the most suitable approach to improve fraud detection performance. First, the model is trained on the original dataset without additional data balancing. When no additional data balancing is applied, the model exhibits a very high overall accuracy since the dataset contains a very high proportion of legitimate transactions. But this high accuracy is deceptive as we notice the model does a very poor job of discriminating the fraudulent transactions from the legitimate ones, since it has a very low recall for the fraud class. This indicates that the model is strongly biased towards predicting each transaction to be legitimate and is not effectively learning the fraudulent behaviour patterns. Therefore, we chose to apply a GAN based approach to generate additional fake fraud samples and balance the dataset distribution.

We observe that the model becomes significantly more capable of identifying fraudulent transactions, which improves the recall. However, we also observe that the false positives also go up slightly and the overall accuracy takes a hit to make this possible. To prevent this problem, we devised a CGAN based approach which generates realistic and condition specific fraudulent samples to help the model strike a better balance without leading to an increased number of false positives. We reckon that a detailed comparison of these methods will help determine the optimum method of constructing a reliable and powerful fraud detection system, as shown in Figure 5. As can be seen from the figure Figure 6, it helps the model to learn and recognize the key features of a fraud business and improves the rate of detecting fraud transactions. The system is working well and it is giving reliable results under this method. This is the performance figure of the proposed fraud detection system and it shows here using accuracy, precision, recall and F1-score measure. It gives a good comparison of how successfully this model helps to locate fraud and normal transactions. And the figure shows the contribution of this model using the CGAN-based approach. Overall, it helps to bring the introduced fraud detection system to understand its productivity and dependability. Overall the experimental of the above shows that, just having accuracy as the only indicator is not sufficient for a fraud detection task. A model trained using an imbalanced data set may have a high accuracy just because most of the transactions are normal, and it will not be able to help you to identify actual fraud cases, though it has a low recall. Although the model with CGAN-financed data training may have lower accuracy overall but it will be helpful to locate real fraud cases. It can perform much better at locating fraudulent transactions and will give very reliable results.

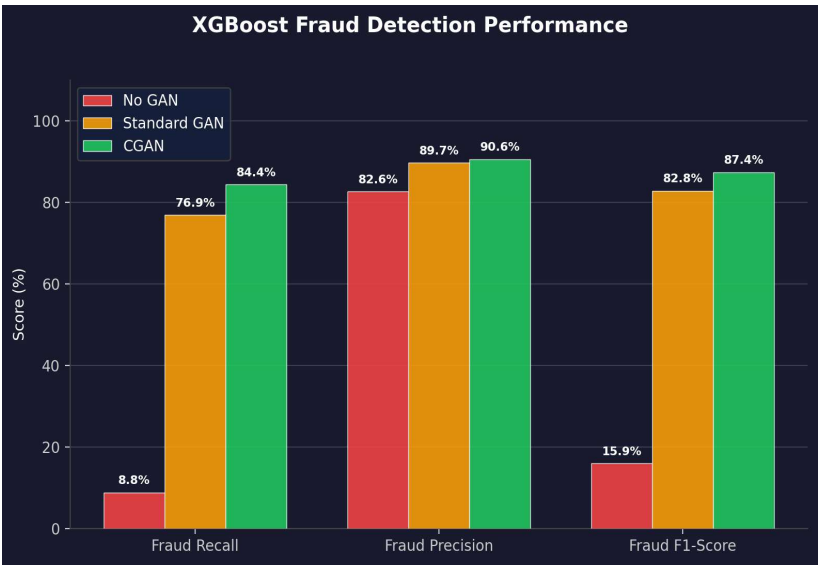


Figure 6: Metrics

Screenshots of results:

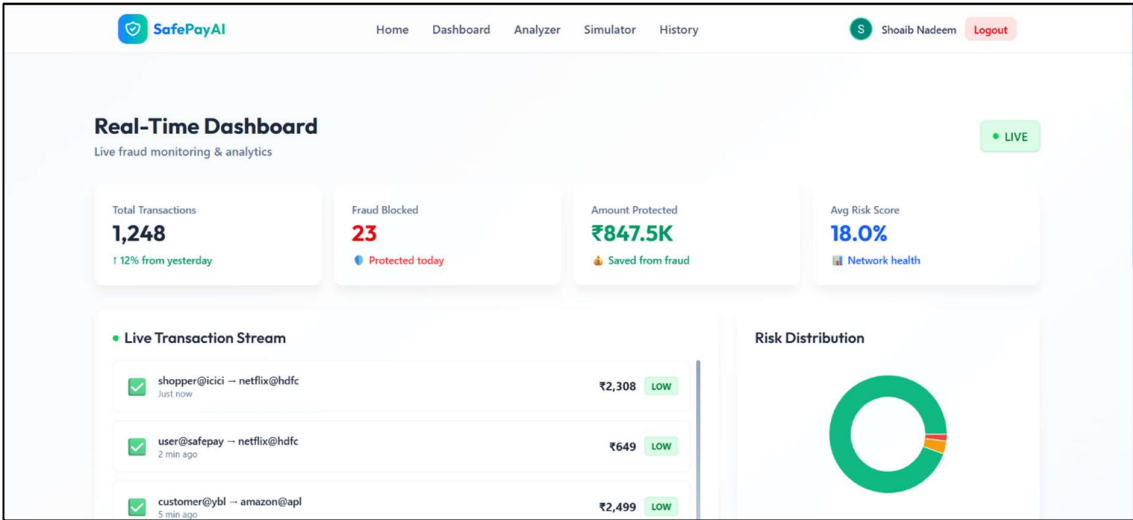


Figure 6: Web Dashboard

The Real-Time Dashboard of the Proposed System, displayed on figure 6, offers a neat, simple and user-friendly graphical presentation of the system’s real time fraud detection activities. It is relatively simple yet very informative, enabling the user to analyze and comprehend the systems current state of application with minimum knowledge of the system itself or the underlying complexities. The first indicators to take into account are the total number of processed transactions, the number of blocked or identified frauds, the total amount of money safeguarded and the average risk score respectively. These indicators collectively offer a profound and concise insight of what is going on in the risk environment. The most valuable element of the dashboard is the stream of real-time transactions which is being constantly updated. These real-time transactions are displayed together with the current risk score. The second element is the wheel chart by displaying visually the number of transactions of each risk level, medium, low or high.

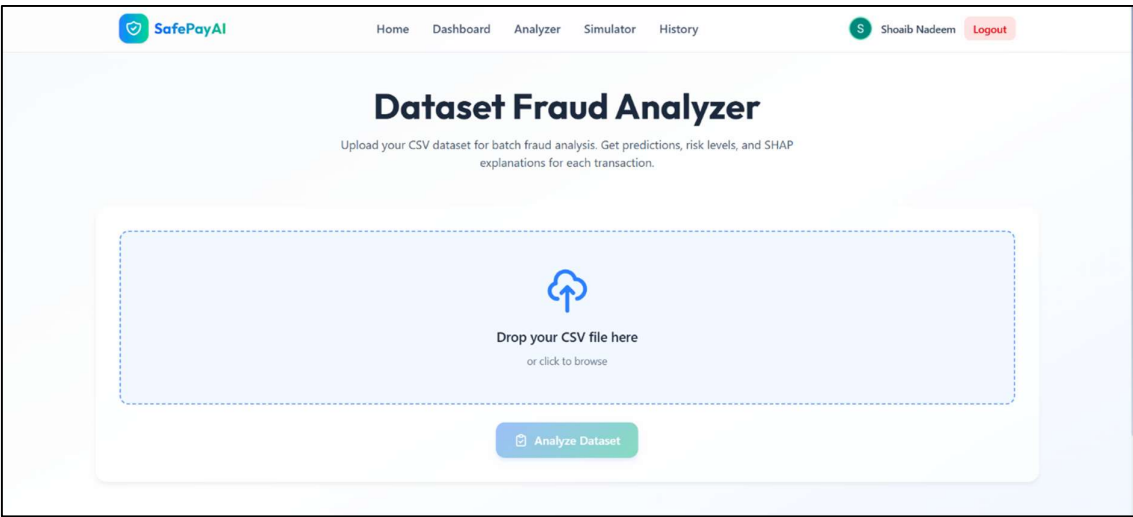


Figure 7: Dataset Analyze

The Dataset Fraud Analyzer in Proposed System as shown in below figure 7 is the fraud detection tool that users can find fraudulent activity faster in huge transaction data. The user interface is simply from choosing the file upload point to start the program. The file can be uploaded using drag-drop or browse it from PC. This will save user easy to find the file. The module came preloaded with the training data and once the data was uploaded, the server used trained machine learning models to process the transactions, identify patterns and find suspicious behaviour. It not only flagged the potentially fraudulent transactions but also did an analysis of its predictions so that bank analysts and financial investigators can understand what is causing the decision.

In the Analysis Results section of the Proposed System, as seen in Figure 7, the output of the fraud detection model is presented in a simple, easy-to-follow way that allows a user to swiftly identify whether the transaction is safe, or otherwise potentially fishy. Each row includes information such as whether the transaction is predicted as fraudulent or legitimate, what the probability of this event occurring is, and the risk score of this fraud event being low, medium, or high. One of the greatest advantages of this section is its effective implementation of explainability. Instead of only returning the fraudulent or legitimate class label, the section reasons what lead to this particular prediction being made. It will identify the top reasons based on the model’s prediction importance, such as acutely high amount, discontinued location, or anomalous user behaviour in the time series.

Analysis Results				
Click on a row to see SHAP explanation				
#	Prediction	Fraud Score	Risk Level	Explanation
13	FRAUD	99.6%	HIGH	Top factor: Transaction Context Anomalies (increasing fraud risk by 2.390)
14	LEGIT	46.4%	LOW	Top factor: Geo-Location Flags (increasing fraud risk by 0.943)
15	FRAUD	99.8%	HIGH	Top factor: Transaction Context Anomalies (increasing fraud risk by 2.875)
16	FRAUD	99.5%	HIGH	Top factor: Time Since Last Transaction (increasing fraud risk by 2.605)
17	FRAUD	76.7%	MEDIUM	Top factor: Geo-Location Flags (increasing fraud risk by 0.848)
18	FRAUD	98.4%	HIGH	Top factor: Fraud Complaints Count (increasing fraud risk by 2.745)

Figure 8: Analysis Results

The UPI Transaction Simulator module in Proposed System (see Figure 8) is a modern, interactive system that simulates a payment transaction to enable users to assess in real time how risky the payment is. It provides a straightforward user interface where the user can identify some of the key parameters for the transaction such as the recipient’s UPI ID and amount, as well as some system-defined default values for sender parameters. Once the form is submitted and the simulation is executed, the system internally propagates the data using the trained machine learning models for detecting fraudulent transactions, and immediately loads the warning or confidence information, as well as the classification label of the transaction simulation, so that the user can determine whether the payment is safe to proceed or not, and search for pattern combinations that cause flagged transactions.

The UPI Transaction Simulator for the Proposed System (see Figure 9) shows the ability of the system to look for potentially fraudulent transactions. When the user submits the transaction details, the system looks for red flags and determines whether the transaction is safe or risky. Here, the system recognizes that this transaction is high risk, and it marks the transaction

accordingly. If users had proceeded with this transaction, they could potentially have lost money from a fraudulent transaction. The simulator also gives a score as to the likelihood of fraud-based on similar past transactions. Here, the score indicates an 88% likelihood of fraud. The simulator allows users to get an idea of the risk involved beforehand as well as better understand the reason for the system marking transactions risky.

Here, the system flags that the payee is a previously cited suspect of fraudulent transactions. As an additional method, the simulator explains using graphics how the transaction is flagged. Here, there are red bars that indicate factors that make the transaction suspicious –such as the location of the transaction is suspicious or the recipient is unverified. There are also green bars that suggest the transaction would be safe –such as the transaction was a small amount or the recipient part of the system.

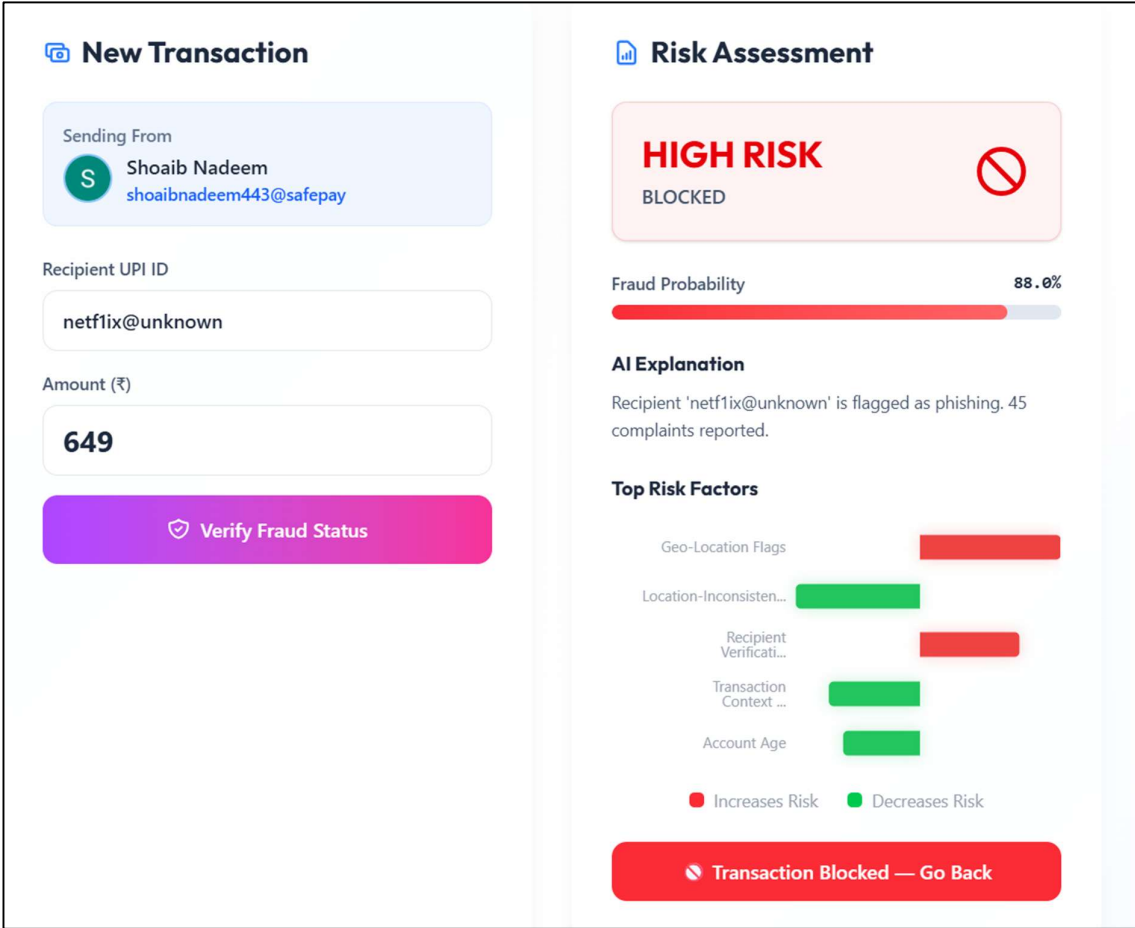


Figure 9: UPI Transaction

Significance of the Proposed Work with its Advantages

Compared to traditional fraud detection processes, the Proposed System is overcoming the three main issues faced by these systems: class imbalance (using CGAN), lack of explainability (using SHapley Additive explanations) and detection power (using the fraud data as a base for both the CGAN and the classifier). Thanks to the use of the CGAN to generate high-quality fake fraud data, this system can present a higher amount of fraud data (helps the learning process) and higher-fidelity training data (using the Conditional information) to both GBT and NG Boost, leading to an overall better prediction performance, especially in the

recall and the F1-score metrics, which are the most important for fraud detection systems. The Proposed System uses the robust GBT XGBoost model and the probabilistic NG Boost model, which provides information about the uncertainty around each prediction. Additionally, the use of the explainability method, SHAP, helps provide transparency over the system's predictions. The Proposed System has lots of advantages for all financial institution, payment platform, and individual user. It brings many advantages to all of them, such as decreasing the losses caused by fraud and increasing security of the transaction. The Interpretability and Accuracy of the system increases trust of the users. It makes the system more reliable. Future work should make the system scalable and flexible.

Additionally, by continuously learning, the system is able to better learn complicated and continuously changing fraud patterns, and is better than traditional rule-based systems at combating fraud. Another feature benefiting the system are probabilistic models to better guide the risk management process with confidence for each default. It is automated, thus minimizing time and cost to banks and payment service providers. Its modular design is easy to implement in existing banking and payment systems. With retraining and new sets of data, the system will continue to evolve in the future to better address fraud issues.

VI. CONCLUSION AND FUTURE ENHANCEMENTS

The emergence of digital payment systems such as Unified Payments Interface (UPI) has revolutionized the way financial transactions are undertaken. People are increasingly using digital payment systems because of the advantages in terms of convenience, fastness and security. However, the rapid adoption of digital payment systems has also increased the incidences of fraud to a considerable extent and has become a huge challenge for the financial institutions and the users at large. Therefore, it has become very important to devise means of quickly detecting fraudulent activity to ensure a more secure and trusted payment system. In this project, an effective fraud detection system will be designed and developed to overcome the drawbacks in other traditional fraud detection systems. A key challenge in fraud detection is highly imbalanced datasets where fraudulent samples can be a very small proportion of the total dataset. Traditional machine learning algorithms apply in such imbalanced cases produce high accuracy yet poor recall due to which fraud detection accuracy is compromised. To prevent this, features such as CGAN with a few other effective algorithms like XGBoost, NGboost and SHAP, etc will be used. The CGAN model will be used to generate fraudulent samples, similar to real fraudulent samples.

The system is based on XGBoost as the main classification model since it can learn the structure of the data and carry out approximation of highly non-linear relationships within transaction data. This feature makes it highly desirable in the largescale financial scenario. NGBoost is also integrated into the framework to provide probabilistic estimates for each transaction so the system can include confidence levels/risk scores to each decision, making the system decision more risk-sensitive, which is highly desired by the financial systems where mistakes might come at high cost. The explainability system based on SHAP also reduces the black-box problem where the application of the system does not require a user to blindly trust the system output without understanding it. The experiment results show that the proposed approach can efficiently reduce the amount of false positive and false negative transactions, covering the high recall and high precision which is the most critical in fraud detection problems. The XGBoost based model can achieve recalled rate of 84.4% and precision of 90.6% and F1-score of 87.4%. Aside from the high accuracy of the fraud detection, the flexible architecture allows a user to switch on new features and modules in the system pipeline to be included with minimal effort. The general combination of generative approach, machine learning,

probabilistic prediction, and explainable AI to build a scalable architecture for fraud detection system is shown effective in addressing challenges faced by most existing methods such as class-imbalance, low recall and explainability. Overall, the system provides a highly scalable, accurate, safe, and interpretative framework that can be effectively integrated and deployed onto a real-world digital payment environment. Additionally, the project stresses much on choosing the right evaluation metrics for fraud detection application, where accuracy is less emphasized than recall and F1-score since the most important factor is identifying all fraudulent transactions and maximize recall instead of overall accuracy as in most traditional classification problems. The focus on improving the recall is shown to minimize the financial losses when preventing frauds.

VII. REFERENCES

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